Michael Ajao-Olarinoye, PhD, AFHEA

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Personal Profile

I am an AI scientist and engineer working on generative AI and AI agents, physics-informed machine learning for cardiovascular hemodynamics and CFD surrogate modelling, and applied machine learning and data science. I am a research fellow at Coventry University and a lecturer in computing, with a PhD in Computational Science and Mathematical Modelling and UK Global Talent endorsement from the Royal Academy of Engineering (Exceptional Promise). Through miolajtech I build AI systems and complete automated business systems for organisations in the UK, Nigeria and internationally, and I tutor students, researchers and professionals in machine learning, data science and scientific computing. My work spans the full lifecycle, from problem formulation and model design to deployment, publication and student supervision, in interdisciplinary and international collaborations.

Research Expertise & Technical Skills

- **Scientific ML & Modelling:** Time-series and spatiotemporal forecasting, physics-informed neural networks (PINNs), hybrid models combining mechanistic (ODE/compartmental) and neural approaches, graph neural networks, statistical modelling, optimisation
- **Machine Learning & AI:** Generative AI, large language models (LLMs) and AI agents, retrieval-augmented generation, deep learning, natural language processing, computer vision, reinforcement learning; PyTorch, TensorFlow, Keras, Scikit-learn
- **Programming & Data:** Python, R, Julia, SQL; Pandas, NumPy, Matplotlib, Seaborn, Plotly; ETL pipelines, data visualisation, classification, regression, clustering, anomaly detection
- **Research & Professional:** Research design, algorithm development, academic writing, peer review, literature review, student supervision, project management, scientific presentation
- **Software, Cloud & Tools:** AWS, Google Cloud, Microsoft Azure; Docker, Git (GitHub, GitLab); web and mobile development (JavaScript, React, Flask, Flutter); LaTeX

Education

- **Coventry University** — *May 2021 – February 2026*
  Doctor of Philosophy (PhD) in Computational Science and Mathematical Modelling (awarded) — Coventry, United Kingdom
  - Thesis: Data-Driven Algorithms for Forecasting Critical Care Demand during the COVID-19 Pandemic, completed through Coventry's Centre for Data Science.
- **Coventry University** — *January 2020 – January 2021*
  Master of Science (MSc) in Data Science and Computational Intelligence — Coventry, United Kingdom
  - Dissertation: Plant Disease Detection: A Cross-Platform Mobile Application Using Artificial Neural Networks, achieving a Distinction
  - Key Modules: Machine Learning, Advanced Machine Learning, Deep Learning, Artificial Neural Networks, Big Data Management and Data Visualisation, and Statistical Methods for Data Science.
- **Kwara State University (KWASU)** — *September 2011 – July 2016*
  Bachelor of Engineering (BEng) in Electrical and Electronics Engineering — Malete, Nigeria
  - Project: Designed and built a 2km Low-Power FM Transmitter, achieving a Second Class Honours
  - Key Modules: Engineering Mathematics, Modelling and Simulation, Digital Signal Processing, Control Engineering, Artificial Intelligence and Application, and Applied Computer Programming.

Work Experience

- **Coventry University** — *January 2026 – Present*
  Research Fellow, Centre for Computational Science & Mathematical Modelling (CSMM) — Coventry, United Kingdom
  - Lead and support research on the Innovate UK-funded HOPE-MOVE project under Professor Matthew England, including literature reviews, study design, fieldwork, and analysis of quantitative and qualitative outcomes.
  - Design and manage machine learning and generative AI applications, ensuring robust data quality, ethical AI assurance, and responsible research practice throughout the project lifecycle.
  - Develop research objectives, methodologies, and proposals for individual and collaborative projects, contributing to the research portfolio and future funding opportunities.
  - Publish research outputs, communicate findings to academic and non-academic audiences, and support student supervision and research-informed teaching.
- **QAHE Ltd** — *February 2025 – Present*
  Lecturer in Computing — Birmingham, United Kingdom
  - Delivered high-quality teaching in computing modules including Programming, Data Science and Machine Learning at Ulster University and Southampton Solent University through QAHE's university partnerships.

- Supervised more than 10 MSc dissertations at Ulster University, providing research guidance, methodological support, and feedback through to thesis completion.
- Designed and implemented course materials, assessments, and interactive learning activities aligned with university curricula and learning outcomes.
- Provided academic support and mentorship to diverse student cohorts, employing inclusive teaching strategies to accommodate various learning styles.
- Collaborated with academic teams across partner universities to maintain teaching quality and assessment standards, and participated in curriculum development and quality assurance processes.

- **Coventry University** *April 2021 – January 2025*
  Postgraduate Researcher Coventry, United Kingdom
  - Developed data-driven forecasting models for COVID-19 hospitalisation and intensive care demand in England, including comparative deep learning architectures, a hybrid physics-informed neural network–SEIRD model, and a multi-scale temporal graph attention network for spatiotemporal epidemic forecasting.
  - Authored peer-reviewed publications and presented research at prestigious conferences, including ICMLA, significantly contributing to institutional research output.
  - Established and sustained an international research collaboration with the National University of Sciences and Technology (NUST), Pakistan, on physics-informed neural networks for cardiovascular hemodynamics, resulting in joint publications in *Physics of Fluids*.
  - Mentored postgraduate students and collaborated with interdisciplinary teams on AI-driven and healthcare projects.
  - Served on the Program Committee for the 23rd IEEE International Conference on Machine Learning and Applications (ICMLA), contributing to peer review and conference organisation.
  - Actively participated in internal conferences and academic workshops, fostering an interdisciplinary research environment and contributing to department-wide initiatives.

- **The Futureworks (Coventry University Group)** *April 2021 – January 2025*
  Graduate Teaching Assistant (Part-time) Coventry, United Kingdom
  - Developed and delivered tutorials, workshops, and individual student support in the School of Electrical, Electronics and Computing, including on advanced modules such as Information Retrieval, Big Data Analytics, Deep Learning, and Computer Vision.
  - Marked and graded assessments, and provided constructive feedback to students.
  - Continuously sought feedback from peers and professors on my teaching to improve my practices and ensure high-quality output.

- **Tammy Tech** *March 2020 – April 2021*
  Software Developer Lagos, Nigeria
  - Designed and developed web and mobile applications using technologies such as HTML, CSS, JavaScript, and Flutter, enhancing client product offerings by 30%.
  - Collaborated with clients to translate their needs into robust technical solutions, applying agile methodologies to deliver projects on time.
  - Integrated various APIs and built databases to support mobile app functionality, improving user experience and engagement.
  - Kept abreast of the latest programming trends and practices through continuous professional development, applying them to project work.

- **Verificar Compliance** *September 2018 – January 2020*
  Business Systems Analyst Lagos, Nigeria
  - Conducted business analysis and developed IT solutions to streamline company processes, resulting in a 15% improvement in operational efficiency.
  - Designed and implemented dashboards and reporting systems to track performance metrics, facilitating data-driven decision-making.
  - Coordinated between technical teams and business stakeholders, ensuring that solutions were aligned with organisational needs.
  - Analysed market trends and emerging technologies to inform product development strategies, positioning the company for future growth.

PUBLICATIONS

- Ur Rehman, M. A., Ekici, Ö., Erdener, Ş. E., **Ajao-Olarinoye, M.**, Kuchumov, A. G., & Jia, F. (2026). Wall shear stress in coronary arteries and grafts using computational fluid dynamics and physics-informed neural networks. *Physics of Fluids*, 38(9), 091905. doi:10.1063/5.0335500
- Ur Rehman, M. A., Ekici, Ö., Farooq, M. A., Butt, K., **Ajao-Olarinoye, M.**, Wang, Z., & Liu, H. (2025). Fluid–structure interaction analysis of pulsatile flow in arterial aneurysms with physics-informed neural networks and computational fluid dynamics. *Physics of Fluids*, 37(3), 031913. doi:10.1063/5.0259296

- Fatima, N., Hamza, M., Farooq, M. A., & **Ajao-Olarinoye, M.** (2026). Modeling of double descending thoracic aortic aneurysms using computational fluid dynamics (CFD) and residual-based physics-informed neural networks (ResNets-PINNs). *Pramana – Journal of Physics*. Accepted, in press.
- **Ajao-Olarinoye, M.**, Babatunde, A., Sadeghimanesh, A., He, F., & England, M. (2026). Physics-informed multi-quantile forecasting for risk-averse NHS critical-care surge allocation. *Proceedings of the 25th UK Workshop on Computational Intelligence (UKCI 2026)*, Coventry. Springer. Accepted, in press; presented 10 September 2026.
- **Ajao-Olarinoye, M.**, Palade, V., He, F., Wark, P. A., Mukandavire, Z., & Mousavi, S. (2025). A hybrid physics-informed neural network–SEIRD model for forecasting COVID-19 intensive care unit demand in England. In U. Onyekpe, V. Palade, & M. A. Wani (Eds.), *Recent Advances in Deep Learning Applications* (pp. 231–256). Chapman & Hall/CRC (Taylor & Francis). doi:10.1201/9781003570882-15
- **Ajao-Olarinoye, M.**, Palade, V., Mousavi, S., He, F., & Wark, P. A. (2023). Deep learning based forecasting of COVID-19 hospitalisation in England: A comparative analysis. In *2023 International Conference on Machine Learning and Applications (ICMLA)* (pp. 1344–1349). IEEE. doi:10.1109/ICMLA58977.2023.00203

Manuscripts Under Review

- **Ajao-Olarinoye, M.**, Palade, V., He, F., Wark, P. A., Mousavi, S., & Mukandavire, Z. (2025). MSAGAT-Net: Multi-scale temporal adaptive graph attention for efficient spatiotemporal epidemic forecasting. Preprint, under review. SSRN. doi:10.2139/ssrn.5719497
- Ur Rehman, M. A., Temimi, H., **Ajao-Olarinoye, M.**, Laadhari, A., Riahi, M. K., & Kissami, I. (Under review). Multiphysics fluid–structure interaction modeling and physics-informed neural network surrogate for morphology-dependent stress localization in thoracoabdominal aneurysms.
- Zahra, **Ajao-Olarinoye, M.**, Farooq, M. A., & Khan, F. (Submitted). Comparative analysis of numerical schemes and physics-informed neural networks for two-dimensional dam-break problems with source terms under different initial dam breaking conditions.
- Jonah, S., Moses, Q., Babatunde, A., **Ajao-Olarinoye, M.**, & Bammeke, D. (2026). LLMs for agentic home energy management. Preprint. arXiv:2607.04569
- Jonah, S., Akinseloyin, O., **Ajao-Olarinoye, M.**, & Babatunde, A. (2026). Controlled evaluation of graph and multimodal augmentation in RAG for document question answering. Preprint. arXiv:2607.16604

Talks and Posters

- **Oral presentation**, 25th UK Workshop on Computational Intelligence (UKCI 2026), Coventry, 10 September 2026. Talk title: *Physics-Informed Multi-Quantile Forecasting for Risk-Averse NHS Critical-Care Surge Allocation*.
- **Invited speaker**, 3rd edition of "Future Algorithms", online, 29–30 April 2026. Talk title: *Algorithmic Training Strategies for Physics-Informed Neural Network Wall Shear Stress Prediction in Cardiovascular Hemodynamics*.
- *Physics-Informed Neural Networks for Fluid-Structure Interaction Analysis of Arterial Aneurysms*, poster presented at the 2nd International Conference of "Future Algorithms", online, 11–12 June 2025 (**First Prize**, poster competition).
- **Talk**, 2nd CSMM Annual Conference, Coventry University, May 2024: *Physics-Informed Neural Networks for Modelling Infectious Disease Dynamics: A Case Study of COVID-19 in England*.
- **Research seminar**, Centre for Computational Science & Mathematical Modelling (CSMM), Coventry University, delivered to colleagues across CSMM and other Coventry University research centres. Talk title: *Data-Driven Algorithms for Forecasting Critical-Care Demand during the COVID-19 Pandemic*, covering deep learning baselines for hospital admissions, a hybrid PINN–SEIR dynamic model, and the MSAGAT-Net graph-based approach for regional multi-step forecasting.
- Presented research at the 1st CSMM Annual Conference, Coventry University (2023), contributing to interdisciplinary exchange on computational science, mathematical modelling, and AI applications.
- Poster presentation at the 2023 IEEE International Conference on Machine Learning and Applications (ICMLA), based on my paper *Deep Learning Based Forecasting of COVID-19 Hospitalisation in England: A Comparative Analysis*.
- *Allocating Medical Resources (Vaccine) to Control a Pandemic*, poster presented at the Coventry University Doctoral Capability and Development Conference (DCAD22), April 2022.

Acknowledged Research Contributions

- Acknowledged for technical discussion that improved Physics-Informed Neural Network (PINN) results in: Abaid Ur Rehman, M., Ekici, O., Farooq, M. A., Talha, R. M., & Amir, S. (2025). *Application of computational fluid dynamics and physics informed neural networks in predicting rupture risk of thoracoabdominal aneurysms with fluid-structure interaction analysis*. *Chinese Journal of Physics*. [link]

Leadership and Volunteering

- **African Institute for Artificial Intelligence (AI4AI)** — *January 2024 – Present*
  Software Engineering Team Lead, AI Literacy team (Volunteer) — Remote
- **Nigeria Union of Engineering Students Association (NUESA), Kwara State University** — *June 2014 – June 2015*
  Vice President — On-site
- **CodeYourFuture, Birmingham, United Kingdom** — *December 2020 – Present*
  Lecturer — Remote
- **National Youth Service Corps (NYSC), Nigeria** — *November 2016 – October 2017*
  Project Manager for Gender Vanguard CDS — On-site

Certifications

- **Chartered Management Institute (CMI)** - 2021:
  - CMI Level 7 Certificate in Strategic Management and Leadership
  - CMI Level 7 Award in Professional Consulting

Honors and Awards

- **UK Global Talent**
  Endorsed by the Royal Academy of Engineering (Exceptional Promise) — September 2026
- **Coventry University**
  PhD Studentship Award (fully funded) — May 2021
- **CSMM Conference**
  Best Reviewer Award — June 2024
- **2nd International Conference of "Future Algorithms"**
  First Prize, Poster Competition — July 2025
- **IEEE International Conference on Machine Learning and Applications (ICMLA)**
  Programme Committee Member — 2024
- **Kwara State University (KWASU), Entrepreneurship and Innovation Challenge**
  Third Place in Community Development Project — October 2017

Research Funding & Grants

- **HOPE-MOVE** (Innovate UK funded project), Research Fellow contribution, Centre for Computational Science & Mathematical Modelling, Coventry University, January 2026 – Present. Applied machine learning, generative AI, and interdisciplinary computational research delivered under the project's UK-funded innovation remit.
- **Coventry University PhD Studentship Award**, fully funded studentship covering tuition fees and stipend, May 2021 – February 2026, supporting doctoral research on data-driven algorithms for forecasting critical care demand during the COVID-19 pandemic.

Professional Memberships

- Associate Fellow of the Higher Education Academy (AFHEA) - 2024 – Present
- IEEE Student Member - 2021 – Present
- Royal Statistical Society e-Student Member - 2024 – Present
- ORCID - Credential ID: 0000-0003-2763-8293

Additional Information

Languages: English (Professional proficiency), French (Elementary proficiency), Yoruba (Native proficiency)
Interests: Reading, Travelling, Teaching, Mentoring, Community Service, Music, and Sports (Football, athletics), AI Ethics, and Sustainability

References

References available upon request.